Hope Enterprise University College	
Faculty of Informatics	
Computer Science Department	
Course Outline	
Course Title:	**Network and System Administration**
Course Code:	**CS 461**
Credit Hours:	**3**
Contact Hours:	**4**
Lab Hours:	**2**
Prerequisite:	**Data Communication and Computer Network**
Course Category:	**Major**
Year:	**4**
Semester:	**1**
Instructor Name	**Eshetu Tadesse**

Course Description

System and network administration is a crucial skill for IT professionals who are responsible for managing and maintaining the technology infrastructure of organizations. This course is designed to provide students with the knowledge and skills needed to effectively administer systems and networks, including managing user accounts and permissions, configuring network services and protocols, implementing security measures, maintaining system performance, and troubleshooting common issues.

Course Objectives

The course aims to provide students with a comprehensive understanding of the fundamental principles and practices of network and system administration. By the end of the course, students will be able to:

- Explain the basic concepts and principles of network and system administration, including network architecture, protocols, and security measures.
- Develop the skills necessary to configure, manage, and troubleshoot network devices and systems.
- Describe how to build network services to users
- Gain hands-on experience with popular network operating systems and tools used in system administration.
- Gain practical experience in managing network operating systems, configuring network settings, and administering network resources.

- Learn about network and system monitoring techniques to ensure optimal performance and identify potential issues.
- Describe the importance of backup and recovery strategies for network and system administration
- Discuss the basic function of network services
- Describe how to install and configure operating system, Active Directory service, Domain control, DNS, DHCP, WEB, etc..
- Describe how to make network resources secure

Develop the skills necessary to manage network users and groups

Course Contents	Weeks
Chapter 1 Introduction to Network Administration	
1.1. System and Network Administrator 1.2. Roles and Responsibilities of Network Administrator 1.3. Designing the Network 1.4. Setting up the Network 1.5. Maintaining the Network 1.6. Expanding the Network 1.7. Applying technology in an environment 1.8. Ethical issues 1.9. The challenges of system administration	**1 , 2 & 3**
Chapter 2: Operating System 2.1 What is Operating System? 2.2 UNIX-like Systems vs. Windows Systems 2.3 Linux Distributions and UIs 2.4 Linux Operations Review 2.4.1 Linux File Hierarchy Structure 2.4.2 UNIX File and Directory Permissions 2.5 Essential Shell Commands 2.5.1 Basic File Manipulation Commands and Directory Navigation Commands 2.5.2 Advanced File Manipulation Commands (Init, Processes, and Threads) 2.6 Advanced Shell Features 2.7 Introduction to Windows Server Operating System	**4, & 5**

2.8 Difference between Linux and Windows operating System	
Chapter 3: Account and Security Administration, and Access Control 3.1 Account and Security Administration, and Access Control 3.2 User and Group Concepts, and User Private Group Scheme 3.3 Managing Files and Folder Permissions 3.4 Managing Users 3.5 Permissions And Restrictions	**6, 7 & 8**
Midterm Exam	**9**
Chapter 4: Network Management Services 4.1 TCP/IP internetworking concepts 4.2 Understanding TCP/IP protocols and their functions 4.3 TCP/IP addressing and subnetting 4.4 Internet Protocol 4.5 Transport Layer Protocol 4.6 Application Layer protocols 4.7 Configuring TCP/IP network settings 4.8 Network and security management 4.8.1 Why do you need security? 4.8.2 User Security Management 4.8.3 Disk Security Management 4.8.3 Disk Management in a security perspective	**10, 11 & 12**
Chapter 5: Network Services and Managing Network Services 5.1 : Network services 5.1.1 Directory Services 5.1.3 Domain Name Services 5.1.4 File Services 5.1.5 Communication Services 5.1.6 Application Services 5.1.7 Print Services 5.2 Managing Network Services	**13, 14, 15 & 16**
Final Examination	**17**

Teaching and Learning Method:

- Lecture
- Problem solving/inquiry

  Lab sessions

Assessment /Evaluation Scheme

Component		Coverage	Assessment Time	
Quiz		Chapter 1&2	Week 5	
Midterm Exam		**Chapter 1, 2, & 3**	Week 9	
Course Project		Chapter 3, 4 and 5	Week 16	
Final Exam		**All Chapters**		
Total				

Text Book

Principle of Network and System Administration Second edition Mark Burgess

Reference Books

1. MCSA Windows Server 2016 Complete Study Guide Will Panek
2. Red Hat Linux Networking and System Administration Terry Collings & Kurt Wall